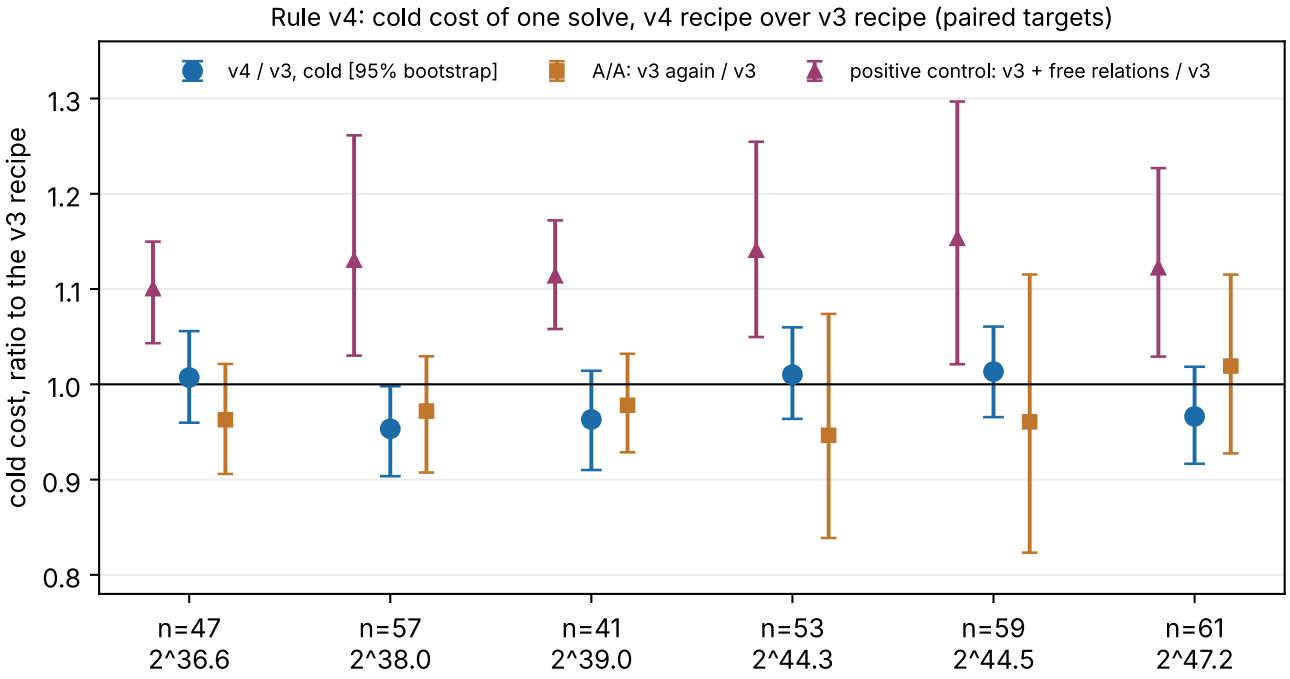
Rule v4: a recipe against v3′s recipe, paired, on the Table A sizes

Outcome: PARTIAL. 1 of 6 sizes improve under the frozen rule (paired cold ratio v4 over v3 with a 95% bootstrap interval below 1); 0 regress. Class: engineering where it improves (S falls, the method is the same). Source: `analysis.json`, `recipe.json`, `training/`, `PROTOCOL.md` (frozen before any timed row; amendments A1 to A7 in it), `replay-certificate.json`. The v4 arm is the v3 recipe with K, the collection window and the unit size retuned on training seeds (lever (ii)); lever (i), free relations from the pair table's duplicate keys, was implemented and measured on the training grid (72 paired runs a size: 5 to 9% **more** cost) and is off. Same binary for both arms; one `isolated_bench` process per row; 16 recipe seeds, 2 targets each; this host has no GFNI.

Paired cold ratio, v4 recipe over v3 recipe

n	log2 r	cold, v4 / v3 [95%]	verdict	set-up [95%]	online [95%]	A/A	pos. ctl
47	36.6	1.007 [0.960, 1.056]	NOT SEPARATED	0.999 [0.951, 1.048]	1.42 [1.13, 1.79]	0.963	1.10
57	38.0	0.953 [0.904, 0.998]	IMPROVEMENT	0.951 [0.901, 0.995]	1.12 [0.99, 1.33]	0.972	1.13
41	39.0	0.963 [0.910, 1.014]	NOT SEPARATED	0.965 [0.912, 1.018]	0.71 [0.44, 0.99]	0.978	1.11
53	44.3	1.010 [0.964, 1.060]	NOT SEPARATED	1.009 [0.961, 1.061]	1.26 [0.98, 1.76]	0.947	1.14
59	44.5	1.013 [0.966, 1.061]	NOT SEPARATED	1.013 [0.965, 1.060]	1.10 [1.00, 1.24]	0.961	1.15
61	47.2	0.966 [0.917, 1.018]	NOT SEPARATED	0.967 [0.918, 1.019]	0.69 [0.51, 0.89]	1.019	1.12

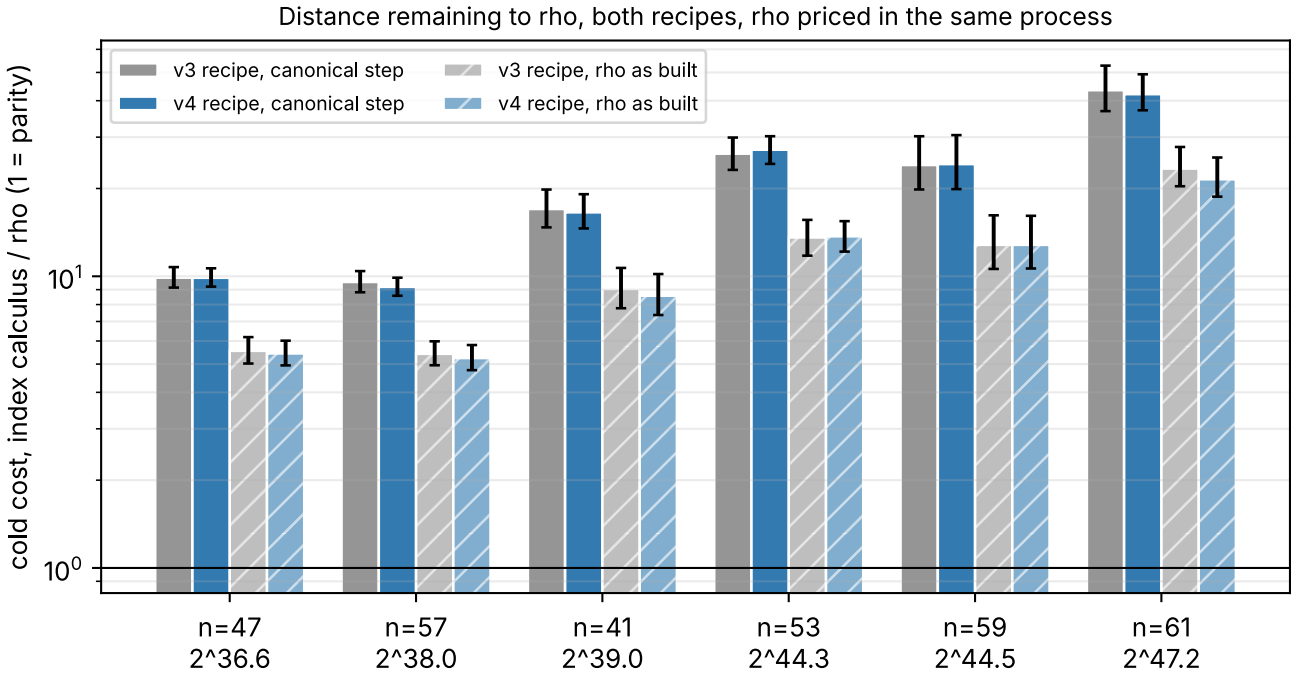
The A/A is the v3 recipe against itself (its interval must contain 1; otherwise the size is UNRESOLVED). The positive control is the v3 recipe with the free-relation lever on (a slowdown of known direction, about 1.05 to 1.09 in the training pairs).



Distance remaining to rho (index calculus cold / rho cold, rho in the same process)

S is cost / sqrt(r) in the process's own batched-addition unit. Canonical step: rho priced at one batched addition plus the table canonicalisation. As built: rho's measured walk. Above 1, rho is cheaper.

n	S_IC v3	S_IC v4	S_rho canonical	gap v3 canon.	gap v4 canon. [95%]	gap v3 as built	gap v4 as built
47	8.85	8.91	0.902	9.89	9.89 [9.22, 10.65]	5.55	5.44
57	8.53	8.13	0.884	9.56	9.20 [8.58, 9.89]	5.42	5.25
41	10.74	10.35	0.624	17.01	16.57 [14.59, 19.12]	9.05	8.58
53	14.73	14.87	0.547	26.33	27.17 [24.29, 30.22]	13.59	13.71
59	14.02	14.21	0.586	24.08	24.25 [19.91, 30.49]	12.82	12.82
61	21.97	21.23	0.504	43.46	42.12 [37.09, 49.28]	23.41	21.52



Section 5a band (counted group-addition equivalents; unpriced native work at 0, 0.1, 1 GAE)

n	gap v4 w=0	w=0.1	w=1	v4/v3 w=0	w=0.1	w=1	order holds
47	17.88	18.84	23.14	1.092	1.101	1.136	no
57	18.30	19.36	24.08	1.004	1.013	1.045	no
41	25.96	27.56	34.62	0.888	0.900	0.939	yes
53	37.01	39.79	52.19	1.079	1.078	1.072	no
59	35.73	38.41	50.43	1.099	1.098	1.094	no
61	46.98	50.40	65.70	0.782	0.779	0.769	yes

Counted GAE is a lower bound: the build's cache misses, the linear algebra and the descent's bookkeeping are not priced, so it can rank two recipes differently from wall time (it prices a larger table's build at one addition a pair). "Order holds" = v4 below v3 at all three weights, interval upper end below 1.

The reference: rho against the strong fixture, T01 to T08, ns per step

ParallelRho is the rho of the `ic price` process. At the second half of n = 41 and at n = 53, 59 and 61 those processes ran with the neighbours' jobs active (amendment A8); the strong fixture rows ran with none, so ParallelRho reads slower against it there than it would. Units: the strong fixture's ns a step over this arm's batched-addition unit.

n	ParallelRho ns/step	strong ns/step	ratio	admissible	strong units/step	ParallelRho units/step	canonical units/step
47	122.9	120.1	1.02	False	8.7	9.2	3.77
57	140.5	116.7	1.20	False	8.5	10.0	4.33
41	120.4	111.8	1.08	False	8.2	8.1	3.75
53	119.0	158.5	0.75	True	11.1	8.2	4.00
59	129.9	141.0	0.92	True	10.3	8.8	4.33
61	142.4	131.5	1.08	False	9.3	8.7	4.39

Lever (i) on the training grid: the same configuration and seed, lever off and on

n	pairs	cost on / off	se (log)	trials on / off, v3 K	build on / off, v3 K
47	72	1.062	0.016	0.911	1.43
57	72	1.058	0.018	0.976	1.41
41	72	1.056	0.019	0.995	1.47
53	72	1.078	0.021	0.964	1.40

59	72	1.088	0.019	0.995	1.47
61	72	1.053	0.023	0.955	1.54

The recipe selected on training seeds (disjoint from the evaluation targets)

v3 / v4 values; the lever is off in the v4 arm (it cost 5 to 9% in the training pairs). Predicted: the fitted cost of the v4 configuration over v3's on the training seeds (it is a prediction, not a result).

n	columns	window	unit trials	predicted v4/v3
47	56 / 48	164 / 262	164 / 164	0.944
57	64 / 54	228 / 365	181 / 181	0.979
41	80 / 96	205 / 328	623 / 623	0.960
53	264 / 224	874 / 874	1014 / 254	0.992
59	256 / 218	944 / 944	935 / 935	0.950
61	320 / 464	1220 / 732	3504 / 876	0.890

Checks

Replay: 493 processes replayed in a separate process with `icprog`'s oracle, 0 failures; 24 untimed re-runs. Status verified.